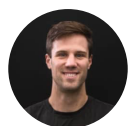


March 27, 2023

Concurrent validity of novel smartphone-based apps monitoring barbell velocity in powerlifting exercises

How accurate can an app be at measuring bar speed? this study compared three commercially available options: Qwik VBT, Metric, and MyLift



Jacob Tober
Co-founder

Great news for strength coaches and athletes using bar speed tracking! A new study by Renner, Mitter, and Baca just dropped in PLOS ONE that puts smartphone VBT apps to the test against high-end lab equipment.

[Check out the full article and PDF at this link →](#)

The research team at the University of Vienna compared three smartphone apps (Qwik VBT, Metric VBT, and MyLift) against **both** a lab-grade 3D motion capture system (Vicon) and a commercially available linear transducer (RepOne). They wanted to see just how accurate and reliable a smartphone app can be at tracking your barbell lifts.

The results?

Qwik VBT — With its video import functionality — was the standout performer, matching or even exceeding the RepOne linear transducer in certain measurements. Our app, Metric v2.3.1, showed solid performance particularly in the bench press, though it had some room for improvement in other lifts (more on how we've upgraded our tracking algorithm since the v2 app used in this study further on in this article). MyLift struggled with repetition detection, missing a substantial number of reps especially in the bench press.

This research is exciting because it confirms what the early-adopters already knew; well-designed smartphone technology can now deliver measurement quality that approaches specialized hardware systems costing hundreds of dollars. The democratization of bar speed tracking technology means more athletes and coaches can access data-driven training methods without investing in expensive equipment.

Let's look at what this study found and what it means for your training approach.

Study Overview

The researchers recruited 20 competitive powerlifters (14 male, 6 female) to put the applications through their paces. Each participant performed a progressive loading protocol for the squat, bench press, and deadlift, working from 45% up to 90% of their recent competition maxes.

For the tech setup three identical iPhone SE 2022 models were mounted on a 1m high custom stand positioned 3m from the end of barbell. The phones ran Qwik VBT (version 0.94), Metric VBT (version 2.3.1), and MyLift (version 3.2.9). Meanwhile, a RepOne tether linear transducer was attached to the barbell sleeve and the entire setup was captured by a Vicon motion tracking system with twelve infrared cameras sampling at 200Hz as the reference.

What makes this study particularly robust is that they collected data across 589 total repetitions with mean velocities ranging from 0.11 to 1.04 m/s, giving a comprehensive picture of how these devices perform across different movements and speeds all performed by lifters with a range of techniques and anthropometries.

Key Findings

Overall Device Performance

When compared to the gold standard Vicon motion capture system, the results showed significant differences between devices:

- **RepOne Linear Transducer** – As expected the LPT performed well with standardized mean bias (SMB) values between 0 and 0.21, and root mean squared error (RMSE) between 0.01 and 0.04 m/s across exercises.

- **Qwik VBT** – was the pick of the smartphone app, with SMB values of 0.00 to 0.13 and RMSE of 0.01 to 0.03 m/s, matching and even exceeding RepOne's performance on some measures. Qwik missed 0 reps throughout the experiment.
- **Metric** – gave mixed results, performing very well on bench press (SMB = 0.07, RMSE = 0.04 m/s), decently on the deadlift (SMB = 0.14, RMSE = 0.08 m/s) but had more variation in squat (SMB = -0.28, RMSE = 0.06 m/s). Metric missed a small number of repetitions on the squat and bench press (3.5 – 6%), and 17% of deadlift repetitions missed.
- **MyLift** – performed similarly to Metric, with SMB values ranging from -0.18 to -0.04 and RMSE between 0.04 and 0.14 m/s. MyLift missed only a very small number of squat and deadlift repetitions (~2%) however it missed 84% of bench press repetitions.

What do these technical statistics mean?

These technical metrics might seem confusing, but they're actually straightforward:

Standardized Mean Bias (SMB) tells you the average bias above or below the true value (truth in this case is the MOCAP values). A number close to zero is better, meaning the device is neither overestimating nor underestimating velocity.

Root Mean Squared Error (RMSE) measures overall accuracy – how close the device's readings are to the true values. Lower is better. For context, an RMSE of 0.04 m/s means that, on average, readings might be off by about 0.04 meters per second.

Repetition Detection Performance

One critical aspect of barbell tracking is the ability to reliably detect all repetitions.

- **RepOne** and **Qwik VBT** both achieved perfect performance with zero missed or ghost repetitions.
- **Metric VBT** missed 52 repetitions out of 500+ (7 in squat, 12 in bench press, 33 in deadlift) and was the only system to register ghost repetitions (16 total: 2 in squat, 14 in bench press – these are likely due to unracking/reracking and can be removed from the data set after a recording).
- **MyLift** struggled significantly with detections, missing 175 repetitions overall with a striking 84% failure rate in the bench press exercise.

The Evolution of Metric VBT

First, congratulations to Qwik VBT on their impressive performance in this study!

Their v0.94 app shows that Metric and MyLift (also available within MyJump) are not the only smartphone velocity tracking options on the market. Since this study was written the Qwik VBT app has improved by adding in-app recording functionality and tracking improvements.

As for Metric, this study tested version 2.3.1, while at the time of writing (April 2025) the newest version of Metric on iOS is 5.2—representing over two years of significant development and improvement. Since the data collection for this study was completed we have been hard at

work addressing the exact issues identified in this research while adding powerful new features.

Major improvements to Metric since v2.3.1:

Tracking accuracy and reliability upgrades:

- Advanced tracking mode specifically designed to support faster repetitions
- Intelligent ghost rep filtering that better identifies unracking and reracking motions
- Enhanced low-light performance for more reliable tracking in poorly lit gym environments
- Processing speed improvements enabling smooth 60fps tracking for greater precision

Feature expansion:

- Real-time audio feedback providing immediate velocity data during sets
- Bar path trace visualization to help athletes improve technique
- Comprehensive workout planner and training log to track progress over time
- Enhanced data export and sharing capabilities

These improvements have paid off. The most recent validation study (Šagovac A, September 2024) testing Metric v4.5 showed excellent correlations with the Vitruve linear transducer in bench press, with correlation coefficients of $r=0.93$ for mean velocity and $r=0.91$ for peak

velocity.

The most notable improvement has been in repetition detection reliability — precisely where version 2.3.1 showed room for improvement in the Renner study. Our latest algorithm is now much more robust at identifying valid repetitions while filtering out false positives, especially in challenging exercises like the deadlift. In the Šagovac study Metric v4.5 detected 100% of the 150 repetitions performed by subjects on the bench press.

While we're proud of how far Metric has come, we view validation studies like this as invaluable feedback for continuous improvement. Each study helps us identify specific areas where we can make Metric even better for our users.

Picking the right bar speed app

The current study provides strong evidence that smartphone-based barbell tracking is a viable tool for coaches and lifters looking to accurately track their training.

This shows that not all smartphone apps perform equally, with significant differences in both measurement accuracy and rep detection reliability. For coaches and athletes, these findings suggest that testing and experimenting with a few properly validated smartphone apps (like Qwik, Metric or MyLift) is a smart starting point before implementing bar speed tracking in your training.

It also highlights the pace and dynamic nature of software, with Metric, Qwik and MyJump all consistently rolling out new features, updates and optimizations in response to feedback and research just like this study. Unlike hardware which remains largely fixed for years at a time,

when reviewing any study looking at applications such as this one, it's always important to check the version number used in the study with [the release notes](#) between when the study was conducted and the most recent app version available on The App Store.

Over on VBTcoach.com, I've curated and continue updating [a comprehensive VBT Buyers Guide](#) listing all the currently available apps, devices and systems for helping you navigate the growing number of bar speed tracking options on the market.

Accuracy and reliability is not the only thing

While this new research offers valuable insights into accuracy and reliability, there are several other factors worth considering.

When selecting a bar speed tracking solution, raw measurement accuracy and reliability is important—but it's just one piece of the puzzle. Consider these additional factors when making your decision:

Ease of use How quickly can you set up and start tracking? Does the app have a steep learning curve, or can you get rolling with minimal friction? The best technology is the one you'll actually use consistently. For example, Qwik VBT delivers great accuracy but still requires manual plate selection for each video, and features like real-time feedback and front-facing camera support aren't yet available.

Support What happens when you have questions or run into issues? Look for apps with comprehensive documentation, a user guide and tutorials, and a responsive support team/person who understand both the technology and training applications. At Metric we have bug reporting features available directly in app, an open Discord, a user guide, the Metric Academy, and an open email policy, you can reach me directly: jacob@metric.coach with your Metric questions at any

time.

Feature completenessBeyond basic velocity tracking, what additional insights does the app provide? Workout planning tools, athlete management plans, progress tracking and more are becoming more and more common across the world of bar speed tracking apps.

Data utilityHow is your training data presented and organized in app? Good visualization and actionable insights can turn raw numbers into meaningful training decisions. Metric offers real-time audible feedback, readiness and fatigue trends, load-velocity, load-power and RPE profiling directly in workout, estimated 1RMs, maximum power tracking, personal record alerts, and workout planning within a single easy to use app.

Development pacesIs the app actively maintained and improved? Technology that evolves based on user feedback and advancing research will provide better long-term value. Checking for an active and steady cadency of improvements, feature releases and bug fixes is a must.

Conclusion

The exciting takeaway from this research is that affordable smartphone solutions really can deliver professional-grade velocity measurement. Whether you choose Qwik VBT, Metric, or another validated app, you're getting access to data that was previously available only through expensive specialized hardware.

This democratization of VBT technology means more athletes and coaches can implement evidence-based, objective training methods — a massive win for strength training as a whole.

As smartphone technology continues to advance expect developers to push these apps to be even more accurate, reliable, and feature-rich. At Metric we are continuing to push and innovate, not only in accuracy of the system but also the powerful and useful ways that velocity tracking data can be utilized in your workouts to deliver what really matters — more gains, fewer injuries and better training.

[← Return to blog home](#)

By lifters for lifters

Metric was built with lifters and coaches front of mind.

Focused analytics and powerful workout tracking tools for Powerlifting, Strength and conditioning, CrossFit, Weightlifting, and anyone serious about their strength training.





METRIC

Barbell velocity tracking

metric app

Metric for
Coaches

Metric User
Guide

Validation

metric data
& metrics

Supported

devices

Blog

Release Notes

support

Privacy policy

Terms &
Conditions

© Core Advantage Pty Ltd. All rights reserved.

